

PRICE LEVEL DETERMINATION IN A MONETARY ECONOMY

FISCAL AND MONETARY POLICY 2026

Piotr Żoch

May 11, 2026

FIAT MONEY NEEDS AN ANCHOR



- Modern currencies are **fiat**: not redeemable for gold, silver, or anything intrinsic.
- A dollar is a promise to deliver another dollar.
- Its value in goods is the price level – itself an **equilibrium object**.
- A theory of inflation is a theory of what gives a dollar its value.

THE PRICE LEVEL PRICES NOMINAL UNITS IN GOODS

- Price level:

$$P_t = \text{dollars per good}, \quad \frac{1}{P_t} = \text{goods per dollar}.$$

- A nominal promise pays dollars, so its real payoff depends on $1/P_t$.
- Real equilibrium pins down quantities and real discount factors, not $1/P_t$.
- A nominal anchor is the extra condition that fixes the goods value of dollars.
- Candidates: interest-rate rules, money demand, fiscal backing.

FISHER AND NOMINAL INDETERMINACY

THE SIMPLEST ECONOMY WITH NO REAL DYNAMICS

$$\max_{\{c_t, \dots\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

- Endowment of nonstorable goods: $y_t = y$.
- Goods market clearing: $c_t = y$.
- P_t = units of account per unit of goods.
- Complete markets: the household trades Arrow securities.
- No government purchases yet.

NOMINAL DISCOUNTING IS A UNIT CONVERSION

- Real stochastic discount factor:

$$M_{t,t+1}^r := \beta \frac{u'(c_{t+1})}{u'(c_t)}.$$

- A real payoff x_{t+1} costs $\mathbb{E}_t \left[M_{t,t+1}^r x_{t+1} \right]$ goods at date t .
- A nominal payoff z_{t+1} buys z_{t+1}/P_{t+1} goods tomorrow.
- Its date- t nominal price is

$$P_t \mathbb{E}_t \left[M_{t,t+1}^r \frac{z_{t+1}}{P_{t+1}} \right] = \mathbb{E}_t \left[\underbrace{M_{t,t+1}^r \frac{P_t}{P_{t+1}}}_{M_{t,t+1}^n} z_{t+1} \right].$$

THE NOMINAL BOND PRICE IS THE EXPECTED NOMINAL SDF

- Nominal stochastic discount factor:

$$M_{t,t+1}^n = M_{t,t+1}^r \frac{P_t}{P_{t+1}} = \frac{M_{t,t+1}^r}{\Pi_{t+1}}.$$

- The price of one nominal unit delivered tomorrow is

$$q_t^n = \mathbb{E}_t M_{t,t+1}^n.$$

- Define the gross and net nominal rates:

$$R_t^n := \frac{1}{q_t^n}, \quad i_t := R_t^n - 1.$$

IN AN ENDOWMENT ECONOMY, THE REAL RATE IS CONSTANT

- Goods market clearing: $c_t = y$.
- The (real) SDF, $\beta u'(c_{t+1})/u'(c_t)$, is constant and equal to β .
- The gross **real interest rate** is

$$R^r := \frac{1}{\beta}, \quad r := R^r - 1 \simeq \log R^r.$$

- In log-linear equations, r is the net real rate.
- The **nominal SDF** is $M_{t,t+1}^n = \beta \frac{P_t}{P_{t+1}}$.

FISHER EQUATION: THE NOMINAL RATE EQUALS REAL PLUS EXPECTED INFLATION

- Gross inflation:

$$\Pi_{t+1} := \frac{P_{t+1}}{P_t}.$$

- Exact Fisher equation:

$$q_t^n = \frac{1}{R^r} \mathbb{E}_t \frac{1}{\Pi_{t+1}} = \beta \mathbb{E}_t \frac{1}{\Pi_{t+1}}.$$

- Log-linear version:

$$i_t \simeq r + \mathbb{E}_t \pi_{t+1}.$$

We ignore Jensen terms, inflation risk premia, and term premia.

FISHER IS A NO-ARBITRAGE CONDITION

- In log-linear notation, $i_t \simeq r_t + \mathbb{E}_t \pi_{t+1}$.
- The household must be indifferent between real and nominal claims.
- This is one equation – it does not by itself pin down P_t .

ONE EQUATION, INFINITELY MANY PRICE PATHS

- In this endowment economy

$$q_t^n = \beta \mathbb{E}_t \frac{1}{\Pi_{t+1}}$$

is the **only** equilibrium condition – we already used goods market clearing.

- What does it imply for the sequence $\{P_t\}_{t=0}^{\infty}$?
- The price path is **determinate** if:
 1. There is a **unique** scalar P_0 in equilibrium.
 2. If Π' and Π'' both satisfy equilibrium conditions, then $\Pi' = \Pi''$.

INTEREST-RATE RULES

DETERMINACY IS ABOUT WHETHER POLICY CAN CONTROL PRICES

- The question may seem esoteric, but it is the heart of the matter.
- It boils down to: **can some policy control prices?**
- In this simple economy, the only available policy is the choice of i_t .
- Suppose we have a target for P_t (or π_t) – can we hit it by setting interest rates?

AN INTEREST-RATE PEG LEAVES INFLATION FREE

- Consider the simplest policy: **interest rate peg**.
- The central bank sets $R_t^n = R^n$ for all t – not a response, a **commitment** in all states.
- The Fisher equation gives

$$\frac{1}{R^n} = \beta \mathbb{E}_t \frac{1}{\Pi_{t+1}}.$$

- This pins down $\mathbb{E}_t \frac{1}{\Pi_{t+1}} = \frac{R^r}{R^n}$ – expected **inverse** inflation.
- It does **not** pin down Π_{t+1} – **actual** inflation:

$$\frac{1}{\Pi_{t+1}} = \mathbb{E}_t \frac{1}{\Pi_{t+1}} + \epsilon_{t+1}, \quad \mathbb{E}_t \epsilon_{t+1} = 0.$$

EVEN UNDER PERFECT FORESIGHT, P_0 IS FREE

- With perfect foresight, $\epsilon_{t+1} = 0$.
- For all $t \geq 0$, Π_{t+1} is determined by the peg, but P_0 is not.
- Fisher pins down growth from P_t to P_{t+1} , not the initial scale P_0 .
- Conclusion: an interest-rate peg does not ensure price-level determinacy.

AN INFLATION PATH STILL NEEDS AN INITIAL PRICE LEVEL

$$p_t = p_{-1} + \sum_{j=0}^t \pi_j.$$

- Inflation determinacy means the sequence $\{\pi_t\}$ is unique.
- Given the inherited level p_{-1} , that sequence implies a unique $\{p_t\}$.
- If the initial nominal scale is free, $p_t + \kappa$ has the same inflation path for any constant κ .
- So a rule can determine inflation without choosing the absolute unit of account.

A TAYLOR RULE TURNS FISHER INTO A DIFFERENCE EQUATION

- Linearized Fisher: $i_t \simeq r + \mathbb{E}_t \pi_{t+1}$.
- **Interest-rate rule:** $i_t = r + \phi \pi_t$.
- The rule is contemporaneous: i_t and P_t are jointly determined in the period- t equilibrium.
- Substituting into Fisher:

$$\phi \pi_t = \mathbb{E}_t \pi_{t+1}.$$

FORWARD-SOLVING REQUIRES $\phi > 1$

- Difference equation:

$$\pi_t = \frac{1}{\phi} \mathbb{E}_t \pi_{t+1}.$$

- Solve **forward**:

$$\pi_t = \lim_{T \rightarrow \infty} \phi^{-T} \mathbb{E}_t \pi_{t+T}.$$

- Bounded-path selection:

$$\phi > 1, \quad \sup_T |\mathbb{E}_t \pi_{t+T}| < \infty \quad \Rightarrow \quad \pi_t = 0 \quad \text{for all } t.$$

THE TAYLOR PRINCIPLE NEEDS A BOUNDED-PATH RESTRICTION

- $\phi > 1$ makes nonzero inflation paths explosive going forward.
- Under the bounded-path restriction,

$$\phi > 1, \quad \sup_T |\mathbb{E}_t \pi_{t+T}| < \infty \quad \Rightarrow \quad \pi_t = \lim_{T \rightarrow \infty} \phi^{-T} \mathbb{E}_t \pi_{t+T} = 0.$$

- Inflation is determinate **within the bounded/non-explosive equilibrium class**.
- Given inherited P_{-1} , π_0 is determined, hence so is P_0 .
- $\phi > 1$ is the **Taylor principle**.
- If $0 < \phi < 1$, π_t is **not** determined – **multiple bounded solutions** exist.

SUNSPOTS ENTER AS FORECAST ERRORS

- Let ξ_{t+1} be extrinsic news with

$$\mathbb{E}_t \xi_{t+1} = 0.$$

- The stochastic process

$$\pi_{t+1} = \phi \pi_t + \xi_{t+1}$$

satisfies the equilibrium condition $\mathbb{E}_t \pi_{t+1} = \phi \pi_t$.

- The question is whether the implied stochastic path is bounded.

A ONE-PERIOD SUNSPOT GIVES CONCRETE PATHS

- Start from $\pi_0 = 0$. At $t = 1$, a sunspot raises inflation by one percentage point.

$\phi = 1.5$ (Taylor principle)

$$\begin{aligned}\pi_1 &= 1, \\ \mathbb{E}_1 \pi_2 &= 1.5, \\ \mathbb{E}_1 \pi_3 &= 2.25, \dots\end{aligned}$$

Path explodes – ruled out by boundedness.

$\phi = 0.5$

$$\begin{aligned}\pi_1 &= 1, \\ \mathbb{E}_1 \pi_2 &= 0.5, \\ \mathbb{E}_1 \pi_3 &= 0.25, \dots\end{aligned}$$

Path stays bounded – sunspot survives, indeterminacy.

The Taylor principle rules out the sunspot only after imposing boundedness.

THE TAYLOR INTERCEPT MUST TRACK THE MARKET REAL RATE

- Suppose r_t is not constant and the central bank follows

$$i_t = \bar{r}_t + \pi^* + \phi (\pi_t - \pi^*),$$

where $\bar{r}_t$ is the rule's real-rate intercept.

- Combining with Fisher:

$$\pi_t - \pi^* = \frac{1}{\phi} [r_t - \bar{r}_t + (\mathbb{E}_t[\pi_{t+1}] - \pi^*)]$$

- Target inflation requires $\bar{r}_t = r_t$: the intercept must match the real return in the Fisher equation.

HOLDEN'S RULE CANCELS THE FISHER REAL-RATE TERM

- Clean one-period benchmark:

$$i_t = r_t + \mathbb{E}_t \pi_{t+1}.$$

- For a constant target π^* , Holden's simple rule sets policy relative to the market real rate:

$$i_t = r_t + \pi^* + \phi (\pi_t - \pi^*), \quad \phi > 1.$$

- Subtracting the two equations:

$$\mathbb{E}_t (\pi_{t+1} - \pi^*) = \phi (\pi_t - \pi^*).$$

- Here r_t is the indexed-bond real yield, **not** the natural rate.

HOLDEN MOVES DETERMINACY OUT OF THE PHILLIPS CURVE

Work in deviations from the inflation target:

$$\hat{\pi}_t := \pi_t - \pi^*.$$

The real-rate rule and Fisher equation give

$$\mathbb{E}_t \hat{\pi}_{t+1} = \phi \hat{\pi}_t.$$

With non-explosive selection, $\hat{\pi}_t = 0$. Then a New Keynesian Phillips curve,

$$\hat{\pi}_t = \beta \mathbb{E}_t \hat{\pi}_{t+1} + \kappa x_t + u_t,$$

determines $x_t = -u_t/\kappa$, not inflation.

HOLDEN STILL USES NON-EXPLOSIVE SELECTION

- Within the non-explosive equilibrium class,

$$\pi_t = \pi^* .$$

- Off-target inflation implies an exploding expected inflation path.
- This is the same selection step as in the Taylor-rule argument.
- Holden follows the standard New Keynesian convention: select the non-explosive path.
- Risk, term, and liquidity premia enter as Fisher-equation wedges; the clean cancellation is the benchmark case.

COCHRANE: LOCAL BOUNDEDNESS IS NOT STABILIZATION

Stabilizing story

$$\pi_t \uparrow \Rightarrow i_t \uparrow \Rightarrow r_t \uparrow \Rightarrow y_t \downarrow \Rightarrow \pi_{t+1} \downarrow .$$

This is old-Keynesian logic.

Determinacy story

$$\pi_t \uparrow \Rightarrow i_t \uparrow \Rightarrow \mathbb{E}_t \pi_{t+1} \uparrow .$$

The alternative path explodes.

Cochrane's point: exploding nominal paths still satisfy equilibrium equations unless another condition rules them out.

COCHRANE: THE DETERMINACY COEFFICIENT IS OFF EQUILIBRIUM

- Write deviations from the selected equilibrium:

$$\tilde{i}_t = \phi \tilde{\pi}_t, \quad \mathbb{E}_t \tilde{\pi}_{t+1} = \phi \tilde{\pi}_t.$$

- In the selected equilibrium, $\tilde{\pi}_t = \tilde{i}_t = 0$.
- So the determinacy coefficient ϕ governs what policy would do in other equilibria.
- Regressions using observed equilibrium data identify the in-equilibrium interest-rate process, not automatically the off-equilibrium threat.

A FUTURE POLICY SWITCH IS A DIFFERENT ANCHOR

- Common defense: if inflation explodes, the central bank will switch policy.
- Then the Taylor rule is not the ultimate nominal anchor.
- The anchor is the policy that would replace it.
- Example: a monetary pillar that takes over when inflation leaves the bounded path.
- Next: can money growth provide that anchor?

MONEY GROWTH

MONEY DEMAND CAN ANCHOR OR ACCOMMODATE

- Lowercase = logs; $\pi_{t+1} = p_{t+1} - p_t$. Money demand:

$$m_t - p_t = y - \eta i_t, \quad \eta > 0.$$

- The same equation does different work under different regimes:

m_t fixed $\Rightarrow$ helps determine p_t ,

i_t fixed $\Rightarrow$ determines endogenous m_t .

- Now suppose money grows at a constant rate: $m_t = \bar{m}t$.

MONEY GROWTH CREATES A STABLE FORWARD EQUATION

- Use Fisher, $i_t \simeq r + \mathbb{E}_t \pi_{t+1}$, and define $x_t := p_t - \bar{m}t$. Then

$$x_t = \lambda \mathbb{E}_t x_{t+1} + d, \quad \lambda := \frac{\eta}{1 + \eta} < 1, \quad d := \frac{\eta(r + \bar{m}) - y}{1 + \eta}.$$

- Solving forward:

$$x_t = \sum_{j=0}^{T-1} \lambda^j d + \lambda^T \mathbb{E}_t x_{t+T}.$$

THE MONEY-DEMAND BENCHMARK RULES OUT REAL-BALANCE BUBBLES

- The forward solution needs one more restriction:

$$\lim_{T \rightarrow \infty} \lambda^T \mathbb{E}_t x_{t+T} = 0.$$

- In this benchmark, money is valuable only because it provides liquidity services.
- For any finite opportunity cost, money demand implies finite desired real balances.
- So we rule out a separate bubble component in $m_t - p_t$.
- This is **not** a general theorem: OLG or bubble-asset models can behave differently.
- Hence

$$p_t = \bar{m}t + \eta(r + \bar{m}) - y.$$

- The anchor here is fixed nominal money supply plus money demand.

MONEY GROWTH AND WICKSELLIAN RULES

- Money growth ensures price-level determinacy here.
- This is true even with stationary shocks u_t to money demand.
- Different microfoundations for money demand may give different conclusions.
- The implied nominal rate is

$$i_t = \frac{1}{\eta} (p_t - m_t + y)$$

- It responds to the **price level** – this is an equilibrium outcome.
- A rule that **starts** with a response to the price level would be a **Wicksellian rule**.
- This is a reduced-form money-demand benchmark; different microfoundations or interest-on-reserves regimes can alter money's anchoring role.

FISCAL BACKING

A MONEY INJECTION IS NOT A COMPLETE EXPERIMENT

- Woodford (1995): money demand may hold, but money is not necessarily the anchor.
- The missing piece is the fiscal counterpart.
- **Open-market swap**: more money, fewer bonds

$$\Delta M_t = -\Delta B_t.$$

Total nominal liabilities are unchanged.

- **Transfer**: more money, no offsetting bond retirement

$$\Delta (M_t + B_t) > 0.$$

This changes fiscal backing unless future surpluses rise.

POLICY LABELS DEPEND ON WHAT IS HELD FIXED

Experiment	Policy move	Held fixed
Monetary	change i_t or liability composition	future surpluses
Fiscal	change $\{s_{t+j}\}_{j \geq 0}$	monetary reaction function

Holding monetary policy fixed means holding fixed the rule, not necessarily the realized instrument path.

The same balance-sheet operation can have different price-level effects under different fiscal counterparts.

A STATIC EXAMPLE: BONDS IN THE MORNING, TAXES AT NIGHT

- Simple static economy.
- Morning: bondholders own B_0 zero-coupon bonds, each promising one unit of account (“dollar”).
- The government pays bondholders by printing “money”.
- Evening: the government collects taxes $P_1 T_1$, paid in “money”.
- The world ends.

$B_0 = P_1 T_1$ IS A VALUATION CONDITION IN THIS REGIME

- In equilibrium

$$B_0 = P_1 T_1.$$

- “Money” printed in the morning must be “soaked up” by taxes at the end of the day.
- B_0 is predetermined; we take T_1 as exogenous.
- Active fiscal policy does not adjust T_1 to validate every possible P_1 .
- The valuation/market-clearing condition selects P_1 .

THE STATIC EXAMPLE ALREADY CONTAINS THE FTPL

$$B_0 = P_1 T_1 \quad \Rightarrow \quad P_1 = \frac{B_0}{T_1}.$$

$$P_1 \text{ too low} \quad \Rightarrow \quad \frac{B_0}{P_1} > T_1 \quad \Rightarrow \quad \text{excess nominal wealth} \quad \Rightarrow \quad P_1 \uparrow.$$

- Government debt in excess of fiscal backing acts like net wealth.
- This is the **Fiscal Theory of the Price Level**.

FTPL VS. THE QUANTITY THEORY OF MONEY

- Quantity theory: inflation comes from **all** money exceeding what is needed for transactions or money demand.
- FTPL: the relevant object is the stock of **outside nominal government liabilities** relative to fiscal backing.
- Whether the liability is currency, reserves, or nominal debt is secondary.
- In the example, “printing” was inessential: people could pay taxes directly with bonds.

NOMINAL GOVERNMENT DEBT CAN BE READ LIKE EQUITY

- The government issues the currency and nominal debt that define the price level.
- One useful analogy: treat nominal government liabilities like corporate *equity*.
- Under this interpretation, their real price adjusts to make a valuation equation hold.
- The analogy applies only to **nominal liabilities of the issuer of the unit of account**.
- It does **not** apply to real debt, foreign-currency debt, or private debt.

COCHRANE (2005): IMAGINE CORPORATE STOCK AS THE UNIT OF ACCOUNT

- Suppose corporate stock is the unit of account and medium of exchange.
- N_t^S = shares outstanding, D_{t+j} = real dividends, P_t^S = price level in shares per good.
- Stock valuation:

$$\frac{N_t^S}{P_t^S} = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j D_{t+j}.$$

- $1/P_t^S$ is the price of one share in goods, hence in the denominator.
- FTPL applies the same logic to nominal liabilities and future primary surpluses.
- This is a [valuation equation](#), not a budget constraint.

CURRENCY REFORM = STOCK SPLIT

- Stock split: more shares, same dividends.

$$N_t^S \uparrow, \quad D_{t+j} \text{ unchanged} \quad \Rightarrow \quad \text{share price falls.}$$

- Currency reform: more nominal liabilities, same real surpluses.

$$B_{t-1} \uparrow, \quad s_{t+j} \text{ unchanged} \quad \Rightarrow \quad P_t \uparrow.$$

- Logic applies to nominal government liabilities, not to real or foreign-currency debt.

ACTIVE AND PASSIVE FISCAL POLICY

- The government **could** set $T_1 = B_0/P_1$ for every P_1 , i.e. $T_1 = f(P_1)$.
- Then the price level is **indeterminate**.
- Such a policy is **passive** or **Ricardian**.
- But the government **need not** follow it. Fiscal policy can be **active** or **non-Ricardian**.

DYNAMIC FTPL

THE HOUSEHOLD PROBLEM WITH MONEY AND ONE-PERIOD NOMINAL BONDS

$$\max_{\{c_t, M_t, B_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$$

$$M_{t-1} + B_{t-1} + P_t y = P_t c_t + P_t s_t + M_t + q_t^n B_t.$$

- $q_t^n := (1 + i_t)^{-1}$; s_t is the real primary surplus; endowment economy: $c_t = y$.
- $M_t \geq 0$. Household debt is bounded by the **natural debt limit** (PV of future endowment) – equivalently, a **no-Ponzi** condition.
- **Transversality** at the optimum: $\lim_{T \rightarrow \infty} \beta^T \mathbb{E}_t(B_{t+T} + M_{t+T})/P_{t+T} = 0$.

IN THIS CASHLESS BENCHMARK, NEW MONEY IS DOMINATED BY BONDS

$$q_t^n < 1 \Rightarrow M_t = 0,$$

$$q_t^n = 1 \Rightarrow M_t \text{ and } B_t \text{ are perfect substitutes,}$$

$$q_t^n > 1 \Rightarrow \text{not an equilibrium.}$$

Setting $M_t = 0$ in the budget constraint:

$$B_{t-1} = P_t s_t + q_t^n B_t.$$

- B_t stands for end-of-period nominal government liabilities (total = bonds in this benchmark).

ITERATING FORWARD GIVES DEBT VALUATION

$$q_t^n = \beta \mathbb{E}_t \frac{P_t}{P_{t+1}} \quad \Rightarrow \quad \frac{B_{t-1}}{P_t} = s_t + \beta \mathbb{E}_t \frac{B_t}{P_{t+1}}.$$

$$\lim_{T \rightarrow \infty} \beta^T \mathbb{E}_t \frac{B_{t+T-1}}{P_{t+T}} = 0.$$

$$\frac{B_{t-1}}{P_t} = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j s_{t+j}.$$

LOW RETURNS MAKE THE TERMINAL TERM IMPORTANT

$$\frac{B_{t-1}}{P_t} = \mathbb{E}_t \sum_{j=0}^{T-1} \beta^j s_{t+j} + \beta^T \mathbb{E}_t \frac{B_{t+T-1}}{P_{t+T}}.$$

- Basic FTPL sets the last term to zero.
- With growth, this is an assumption about debt relative to the scale of the economy.
- If returns are very low relative to growth, debt can be rolled over for a long time without large surpluses – a bubble.
- Then price-level determination depends on the terminal condition, not only on the finite stream of surpluses.

THE FTPL EQUATION SELECTS P_t ONLY UNDER STRONG ASSUMPTIONS

- B_{t-1} is predetermined and nominal.
- Liabilities are default-free and denominated in domestic currency.
- $PV_t(s)$ is positive and finite.
- Fiscal policy fixes future real surpluses independently of the price-level revaluation of debt.
- The real discount factor is determined independently of nominal debt supply.
- The no-bubble/transversality condition holds.

COMPLETE MARKETS TURN VALUATION INTO P_t

$$PV_t(s) := \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j s_{t+j}.$$

- Real discounting is determined by real fundamentals, **independently of P_t** .
- Active fiscal policy fixes $\{s_{t+j}\}_{j=0}^{\infty}$ **independently of P_t** .

$$P_t = \frac{B_{t-1}}{PV_t(s)}.$$

FTPL IS AN ASSET-PRICING EQUATION

$$\underbrace{B_{t-1}}_{\text{nominal claims}} \times \underbrace{\frac{1}{P_t}}_{\text{goods per claim}} = \underbrace{PV_t(s)}_{\text{real fiscal dividends}} .$$

- The price of nominal liabilities adjusts like an equity price.
- This is a valuation equation, not a conventional budget constraint.
- This is the complete-markets active-fiscal benchmark.

SURPLUS NEWS IS INFLATION NEWS

$$PV_{t+1}(s) := \mathbb{E}_{t+1} \sum_{j=0}^{\infty} \beta^j s_{t+j+1}, \quad \frac{B_t}{P_{t+1}} = PV_{t+1}(s).$$

Take the surprise relative to date t :

$$\frac{B_t}{P_t} \left(\frac{P_t}{P_{t+1}} - \mathbb{E}_t \frac{P_t}{P_{t+1}} \right) = PV_{t+1}(s) - \mathbb{E}_t PV_{t+1}(s).$$

BAD FISCAL NEWS RAISES UNEXPECTED INFLATION

$$PV_{t+1}(s) - \mathbb{E}_t PV_{t+1}(s) < 0 \quad \Rightarrow \quad \frac{P_t}{P_{t+1}} \downarrow \quad \Rightarrow \quad \Pi_{t+1} \uparrow .$$

- At time $t + 1$, both B_t and P_t are predetermined.
- With one-period debt, the fiscal-news shock affects inflation for one period.
- With long-maturity nominal debt, fiscal news can also move bond prices and expected future inflation.

LIABILITY-ISSUANCE REVENUE IS PRICED BY FUTURE SURPLUSES

- How much real revenue does issuing B_t raise? Note P_t is **already determined** by $B_{t-1}/P_t = PV_t(s)$.
- Take date- t expectations of $B_t/P_{t+1} = PV_{t+1}(s)$:

$$\frac{B_t}{P_t} \mathbb{E}_t \frac{P_t}{P_{t+1}} = \mathbb{E}_t PV_{t+1}(s).$$

- Substitute the bond price $q_t^n = \beta \mathbb{E}_t P_t/P_{t+1}$:

$$\underbrace{\frac{q_t^n B_t}{P_t}}_{\text{real revenue from issuing } B_t} = \beta \mathbb{E}_t PV_{t+1}(s).$$

- Real revenue is pinned by **future surpluses**. Doubling B_t does **not** double revenue – q_t^n falls.

FTPL CAVEAT AND DEMAND THEORY

WHAT CAN BREAK THE SCALAR FTPL EQUATION?

Complete markets

$$\frac{B_{t-1}}{P_t} = PV_t(s; R^r)$$

R^r is determined by real fundamentals
independently of B_{t-1}/P_t .

Incomplete markets

$$R_t^r = \mathcal{R}\left(\frac{B_{t-1}}{P_t}, \Omega_t\right)$$

R_t^r may co-move with real debt supply.

$$\frac{B_{t-1}}{P_t} = PV_t(s; R_t^r, \dots)$$

Then the equation is a locus in (P_t, R_t^r) , not a scalar equation for P_t .

HAGEDORN ANCHORS P_t THROUGH SAFE-ASSET DEMAND

$$\Pi = 1 + g_B = \frac{B_t}{B_{t-1}}, \quad R^r = \frac{1+i}{1+g_B}.$$

Let $S(R^r)$ denote the household **real demand** for safe nominal claims at real return R^r .

$$\frac{B}{P} = S(R^r) \quad \Rightarrow \quad \boxed{P = \frac{B}{S((1+i)/(1+g_B))}}.$$

- Fiscal policy sets nominal debt growth g_B .
- Monetary policy sets the nominal rate i .
- Private demand for safe real assets sets B/P .

SAFE-ASSET DEMAND REFLECTS PRECAUTIONARY SAVING

- Incomplete markets: households face idiosyncratic risk they cannot fully insure.
- Default-free nominal government debt is a natural **precautionary store of value**.
- Demand schedule:

$$\frac{B}{P} = S(R^r),$$

the desired real holdings of safe nominal claims at real return R^r .

- Microfoundations: Aiyagari and Huggett (idiosyncratic risk + borrowing limits); Krishnamurthy–Vissing-Jorgensen (convenience yield on Treasuries).
- Under **complete markets**, safe and risky claims are priced by the same SDF – no separate $S(\cdot)$.

FTPL VS. DEMAND THEORY: WHO PRICES THE DEBT?

FTPL

$$\frac{B}{P} = PV(s)$$

Marginal pricer: the **fiscal authority**, via committed surpluses.

Household takes $PV(s)$ as given; P adjusts.

Demand theory

$$\frac{B}{P} = S(R^r)$$

Marginal pricer: the **precautionary household**, via demand for safe assets.

Government takes $S(\cdot)$ as given; P adjusts.

FTPL needs **active** fiscal policy. Demand theory is consistent with **passive** fiscal policy.

DEMAND THEORY REQUIRES AN ELASTIC $S(\cdot)$

- For P to respond meaningfully to fiscal and monetary shocks, $S(\cdot)$ must be sensitive to R^f .
- If S' is **small**: the price level barely moves – the framework cannot generate the inflation episodes it claims to explain.
- If S' is **very large**: small shifts in R^f produce big price-level swings – empirically counterfactual.
- Evidence: Krishnamurthy–Vissing-Jorgensen find a sizable Treasury convenience yield, but its sensitivity to debt supply at macro frequencies is debated.
- Open question: are convenience yields large enough at macroeconomic debt levels to anchor P_t , or only at the margin?

SUMMARY: WHAT SELECTS P_t ?

NOMINAL ANCHORS

Mechanism	Held fixed	Key equation	Selection issue
Interest-rate rule	$i_t = r + \phi \pi_t$	$\mathbb{E}_t \pi_{t+1} = \phi \pi_t$	bounded path
Real-rate rule	$i_t = r_t + \pi^* + \phi(\pi_t - \pi^*)$	$\mathbb{E}_t(\pi_{t+1} - \pi^*) = \phi(\pi_t - \pi^*)$	observed market r_t
Money growth	m_t	$m_t = p_t + y - \eta i_t$	no money bubble
FTPL	$\{s_{t+j}\}, B_{t-1}$	$B_{t-1}/P_t = PV_t(s)$	active fiscal policy
Demand theory	B, i, g_B	$B/P = S(R^r)$	safe-asset demand